



Scientific collaboration networks in the interdisciplinary field of biodiversity in central Germany

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Introduction

Scientific and complex economic activities are concentrated in urban and metropolitan areas¹

A selective process of spatial proximity between knowledge producing and demanding institutions²

Remnants of a historical divide putting distance between institutions^{3,4}

Multi-institutional collaborations != simple agreements among researchers⁵

Examining effectiveness of institutional scientific coalitions can inform future policies⁴

- [1] Balland, P.-A., Jara-Figueroa, C., Petralia, S. G., Steijn, M. P. A., Rigby, D. L., & Hidalgo, C. A. (2020). Complex economic activities concentrate in large cities. *Nature Human Behaviour*, 4(3), 248–254. <https://doi.org/10.1038/s41562-019-0803-3>
- [2] Rammer, C., Kinne, J., & Blind, K. (2020). Knowledge proximity and firm innovation: A microgeographic analysis for Berlin. *Urban Studies*, 57(5), 996–1014. <https://doi.org/10.1177/0042098018820241>
- [3] Abbasiharofteh, M., & Broekel, T. (2020). Still in the shadow of the wall? The case of the Berlin biotechnology cluster: *Environment and Planning A: Economy and Space*. <https://doi.org/10.1177/0308518X20933904>
- [4] Akbaritabar, A. (2021). A quantitative view of the structure of institutional scientific collaborations using the example of Berlin. *Quantitative Science Studies*, 2(2), 753–777. https://doi.org/10.1162/qss_a_00131
- [5] Shrum, W., Genuth, J., Carlson, W. B., Chompalov, I., & Bijker, W. E. (2007). *Structures of Scientific Collaboration*. MIT Press.

Why this case?

Previous study on the case of Berlin region:

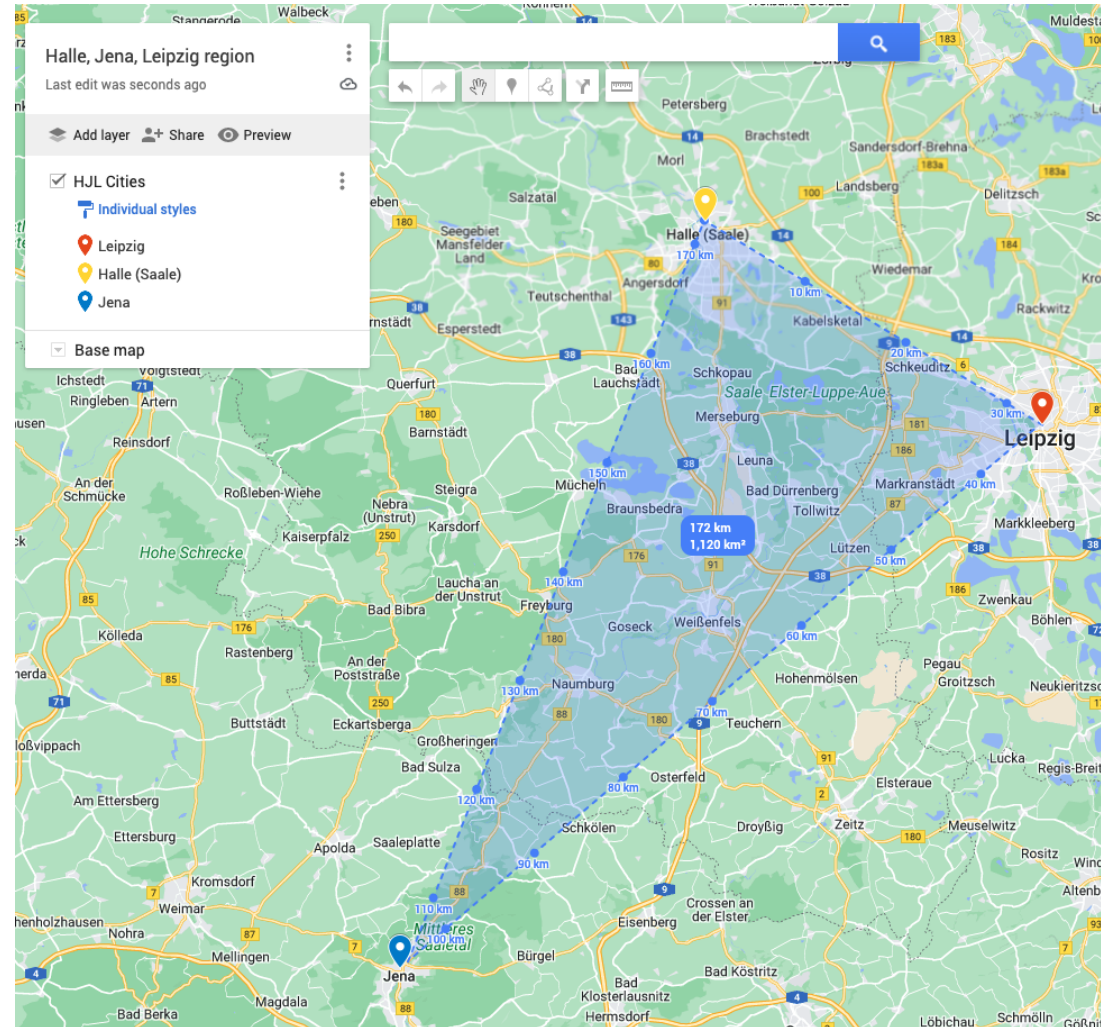
- Divide/competition
- Mutually exclusive identity
- Cooperation

Published in:

https://doi.org/10.1162/qss_a_00131

Halle-Jena-Leipzig
(HJL):

- Not one metropolitan region
- **Unibund** from 1995
- Established **iDiv** with DFG funding in 2012





Questions

- **RQ1:** How **collaborative** and **internationalized** is the scientific landscape of the HJL region?
- **RQ2:** Are there **field differences** in the rate of collaborative and internationalized scientific work?
- **RQ3:** Are there specific field, sectoral, national or continental **cohesive subgroups** driving the scientific collaborations in the HJL region?
- **RQ4:** Can we find evidence of **Unibund's effect** as a **strategic coalition** on the structure of scientific collaborations in the HJL region?
- **RQ5:** Can we find evidence of **iDiv's effect** as a **new organizational form** on the structure of scientific collaborations in the HJL region?



Data & methods

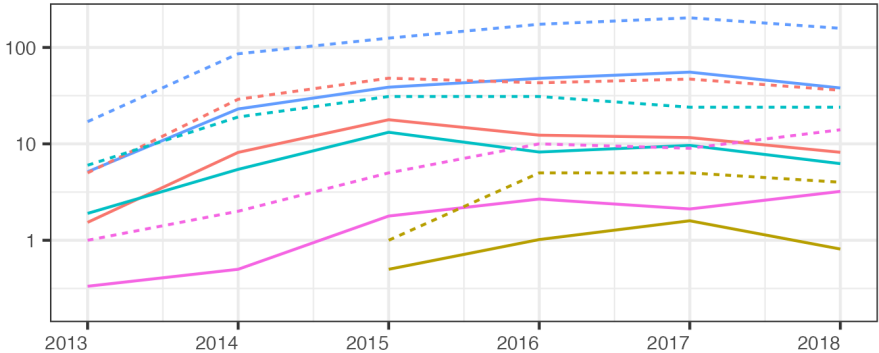
- Scopus **1996-2018** (from German Competence Network for Bibliometrics)
- *Article, Review and Conference proceeding*
- At least one institution of **HJL** (*Halle, Saale, Jena or Leipzig*), **Germany**
- **154,822** publications, **462,426** *Non-disambiguated* institutions
- iDiv website: **30th July 2020**; **2,371** unique DOIs, 1,749 on Scopus 2019 (74%), involving 12,828 *Non-disambiguated* institutions
- OECD fields (mapped to ASJC from Scopus)
- **Wikidata** 27 March 2019, **GRID** (Global Research Identifier Database) February & December 2019
- **Research Organization Registry (ROR)** [local] API (December 2019)
- Bipartite community detection, CPM, $\gamma = 6 \times 10^{-3}$

Multi-institutional collaboration (left), Internationalization iDiv (top right), HJL (right, cities)

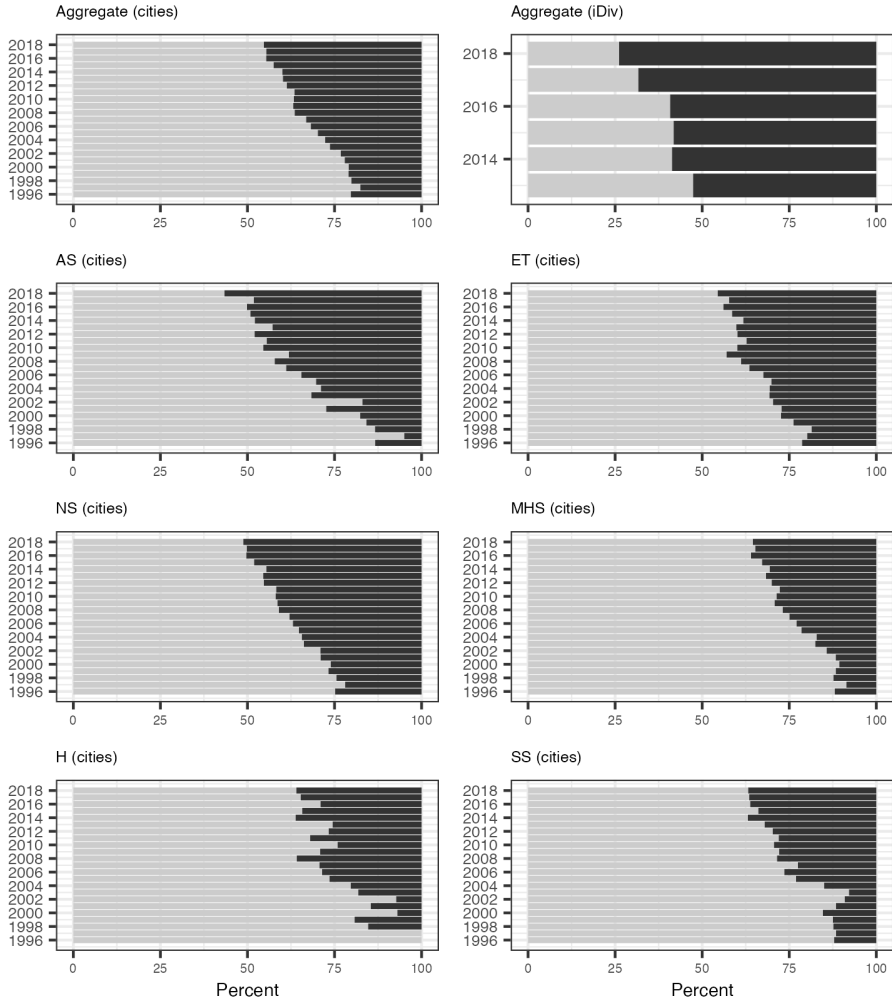
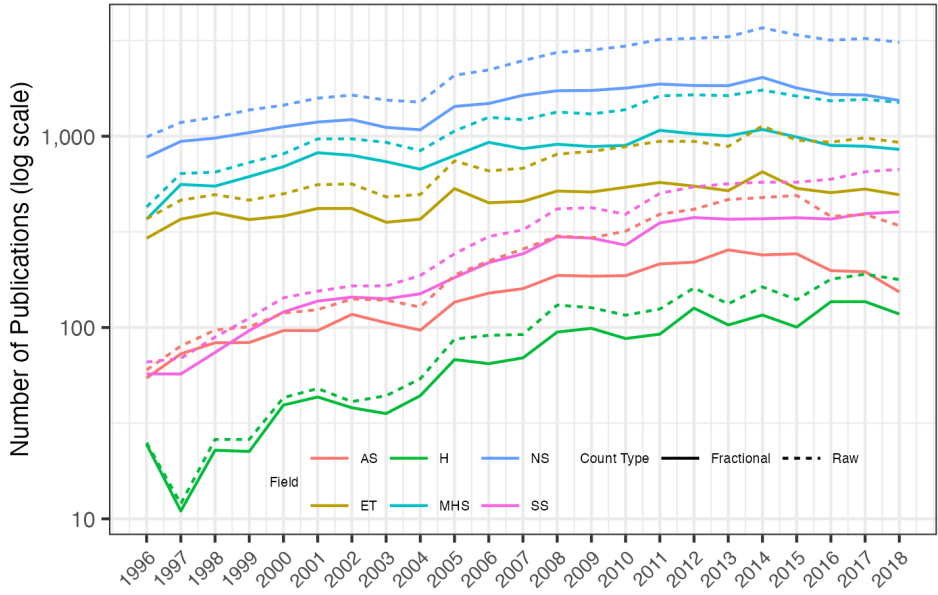
- AS: Agricultural Sciences
- ET: Engineering Technology
- NS: Natural Sciences
- MHS: Medical and Health Sciences
- H: Humanities
- SS: Social Sciences



iDiv

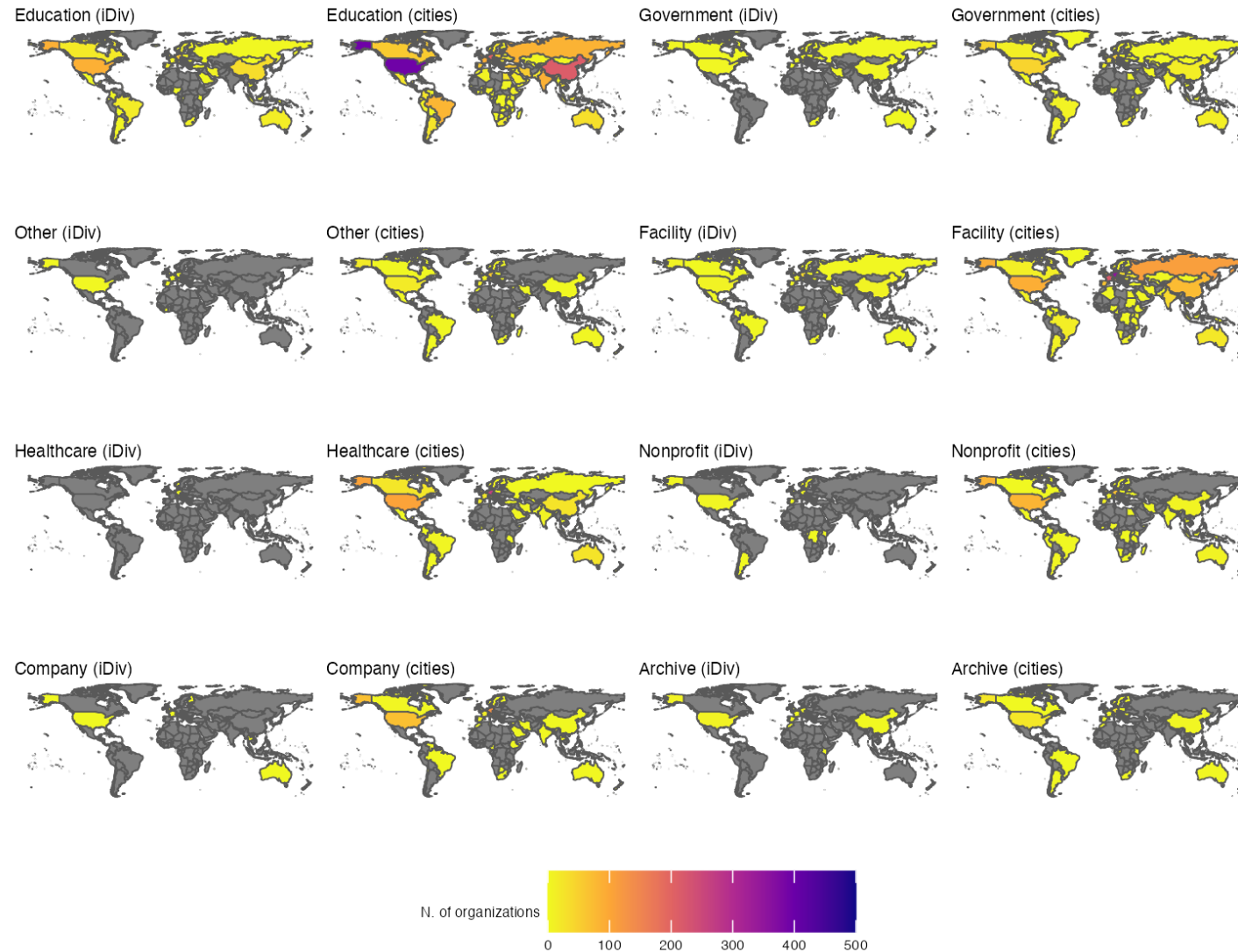


HJL cities



Country Status Multiple Countries pubs Single Country (DEU)

Groupings based on sectors

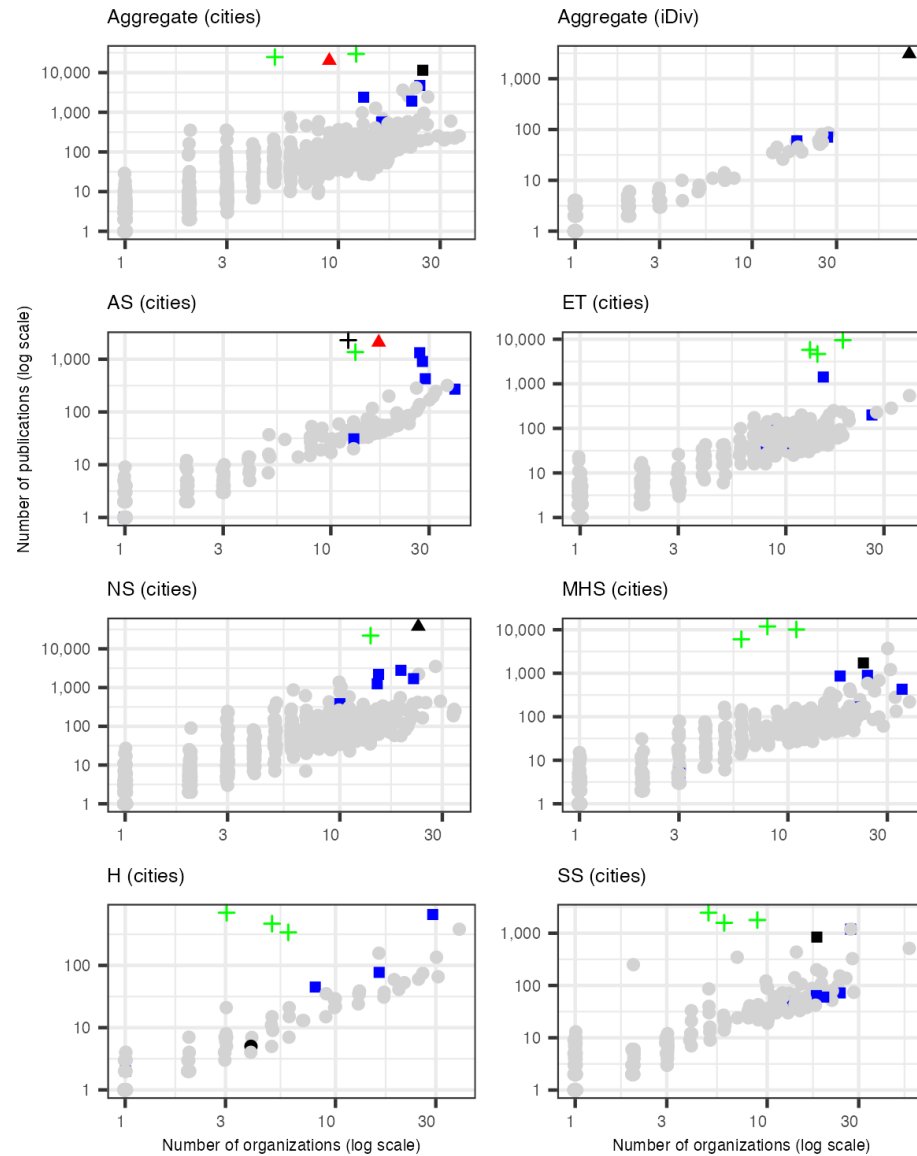




Metrics	Non disambiguated web	Disambiguated web	Non disambiguated cities	Disambiguated cities
N. of connected components	228	1	61,906	22
N. of bipartite nodes	14,577	1,604	617,248	86,233
N. of bipartite edges	16,485	4,312	631,291	168,078
% of bipartite nodes in G	87	100	55	100
% of bipartite edges in G	90	100	65	100
N. of organizations	12,828	686	462,426	6,289
N. of organizations in G	11,243	686	264,938	6,268
N. of publications	1,749	918	154,822	79,944
N. of publications in G	1,462	918	75,055	79,894

Cohesive communities

- Green plus: includes Unibund member(s),
- Blue square: includes iDiv member(s),
- Black: includes iDiv center,
- Triangle/red: includes Unibund/iDiv member(s)





Conclusions & Discussion

- **Internationalization** is increasing, not prevalent yet
- Geographical proximity is not enough for cohesive institutional collaboration
- Low integration of business sector (companies)
- Simultaneous look at **fields**, **sectoral** and **geographical** composition of communities
- Newly established interdisciplinary centre (iDiv) has extensively bridged the member institutions.
- More unrealized potential in the region.
- Level of interdisciplinary collaboration the coalition achieved could inform policymakers



Thank you for your attention

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Community composition (HJL cities (top), iDiv web (bottom))

Data	cluster	N	P	iDiv	I	U	Region (%)								Sector (%)								
							Africa	Americas	Asia	Berlin	DEU	Europe	HJL cities	Oceania	No region	Archive	Company	Education	Facility	Government	Healthcare	Nonprofit	Other
ROR web	0	77	3,061	Y	5	3	3	5	9	4	38	29	10	3		1		70	22	3		3	1
ROR web	1	27	86					30	33		15	11	7	4		4		81	7	4		4	
ROR web	2	27	71		1		4		15		22	59				7		63	7	22			
ROR web	3	25	81				4	28				64	4			12		64	16	4			4
ROR web	4	18	60		1		33	17			11	33	6			6	6	50	11	6		11	11
ROR web	5	24	51					67	21		4	8				12	4	71	12				
ROR web	6	24	53				21	21	17			29		12		4		71	8	4		8	4
ROR web	7	25	57					32	4			52		12		8		64	16	8			4
ROR web	9	24	61					50		4	4	38		4			4	83	4	4		4	
ROR web	12	18	59					61	6		6	17		11		6		78	6			6	6
ROR web	42	2	3		1						100								100				
ROR cities	0	5	24,616			1					20	20		60				60	20		20		
ROR cities	1	12	29,608			1					25	17	50	8				42	42		17		
ROR cities	2	9	20,127		1	1					67		33					67	33				
ROR cities	3	25	11,481	Y	1						48	36	12	4				72	20	4		4	
ROR cities	4	24	4,741		1			42			8	46	4					79	17	4			
ROR cities	7	13	2,398		1			31	15			46	8					62	31	8			
ROR cities	8	26	2,437					19	12		19	46	4					38	46	12		4	
ROR cities	9	22	1,912		1		5	14			27	50	5					27	45	14		9	5
ROR cities	25	16	569		1		6	6	12		44	31					6	56	19	6	12		
ROR cities	36	34	599				12	6	6		9	59		6	3	3	3	29	44	9	3	6	3
ROR cities	49	37	255				5	24	51			16		3			3	68	16	3	8	3	
ROR cities	63	26	491				4	8	19	4	27	35		4				35	23	8	27	4	4
ROR cities	79	35	223					11	49			40				6		60	14		20		
ROR cities	81	29	254				3	3	90			3					3	69	14	7		7	
ROR cities	92	31	200					16	3		6	74				6	3	29	45	3	3	6	3
ROR cities	93	32	221					75	3		6	6		9				38	22	19	9	9	3
ROR cities	108	15	208		1		7		27		27	40						47	33	7		7	7
ROR cities	125	26	209					15	8		4	73				4		46	35	8	4	4	
ROR cities	164	17	109		1			6	6		12	76				18	6	29	35	6			6

N: size of community, P: total publications, iDiv: includes iDiv center, I: number of iDiv members, U: number of Unibund members